FACULTY OF COMPUTING

PROJECT OVERVIEW
SCST1143 DATABASE ENGINEERING

DESCRIPTION

Form a group of **4 members** (or a maximum of 5 members, only if allowed by lecturers).
Your team must identify **ONE system** as a case study to be proposed for your team's project.

The proposed case study should be either a new system or an enhancement of the currently available system. The proposed case study can also be derived from two situations:

a) Manual or not fully computerized/automated systems yet,
b) Existing computerized/automated systems that are having problems.

Your team must identify and analyze the problems and propose solutions to develop new or improve the current system. To complete the project, your team must perform the specified tasks provided in detail for each phase.

PROPOSED SYSTEM REQUIREMENT

Based on the requirements of the stakeholders, each group is required to propose adatabase system.

The proposed system must follow the requirements below:

1. **Fulfilled the project requirement.**
   Please refer to the DB project phases description for further instruction on the report and evaluation for each phase. The evaluation for the project will be based on the course project rubric.

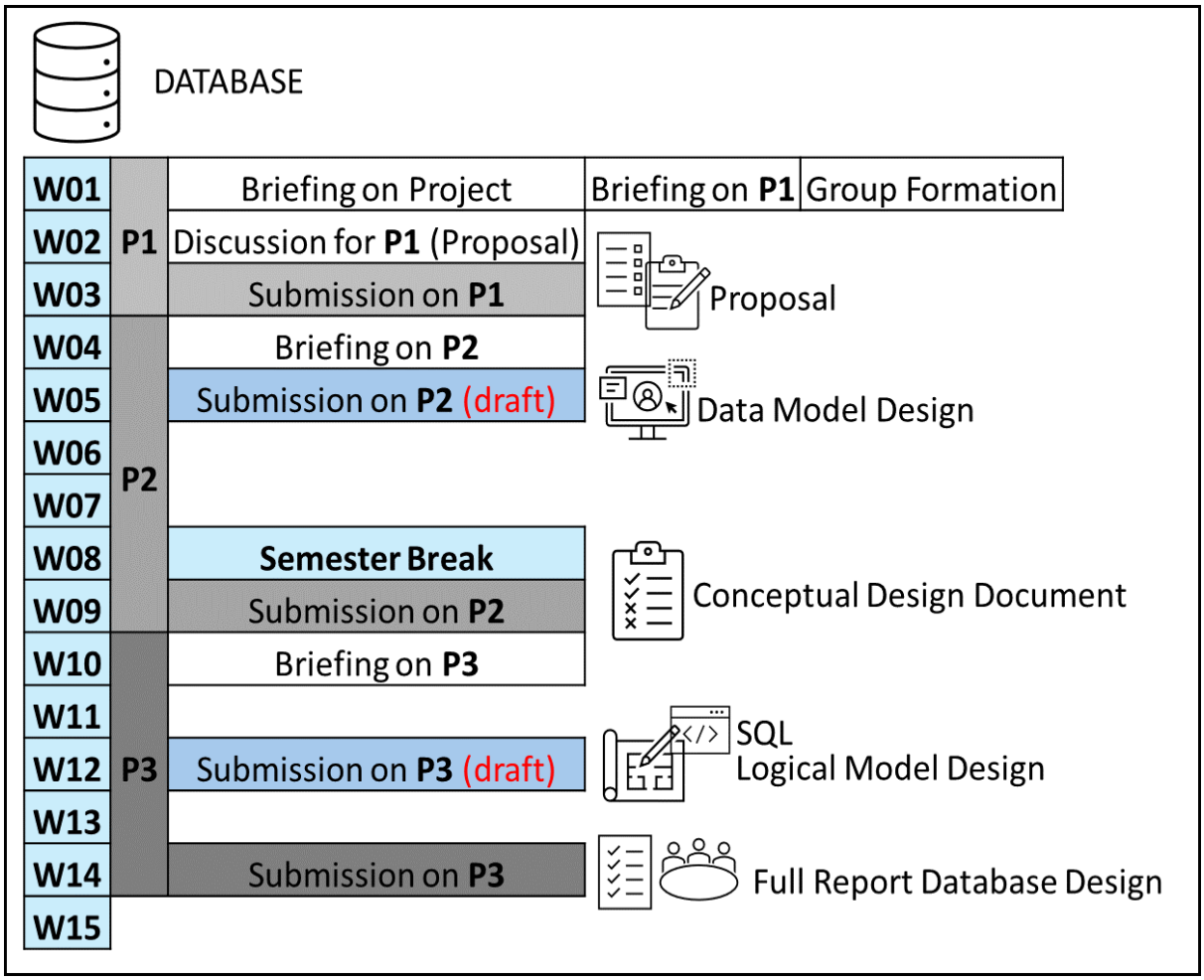
2. **The proposed system must have at least FOUR (4) modules**.
   Each student in the group will be responsible for designing at least **ONE module**. This includes the analysis and design task of designing the module (Process Flow and Entity Relation (ER) design). The module must be designed with consideration of the group's proposed system.

   Each module should be able to be combined/merged to produce the main Process Flow (for the proposed system) and global data model (for the database) – refer figure below.

3. **Students will do a demo of the project in the Final Phase.**
   Students would need to demo:
   - A relational database schema
   - Process flow of the proposed project
   - Working query (DML) based on the DDL design of the project.

PROJECT SUBMISSION TIMELINE

This is just a suggested timeline, please **refer to your section lecturers for further instruction*



PROJECT REPORT REQUIREMENT

	GROUP WORK + INDIVIDUAL		
DATABASE	**P1** **PROPOSAL & DATABASE REQUIREMENT** **[W01 - W03]**	**P2** **DATABASE CONCEPTUAL DESIGN (ERD)** **[W04 - W09]**	**P3** **DATABASE LOGICAL MODEL DESIGN & SQL** **[W10 - W14]**
REPORT CONTENT	**Project Introduction** Overview of the Project Problem Statement Proposed Solutions Objectives Scopes	**from P1** Overview of the Project DFD (to-be) Data & Transaction Requirement Proposed Business Rules Proposed Data and Transactional	**from P2** Overview of the Project Database Conceptual Design Updated Business Rules Conceptual ERD
	System Analysis and Design Feasibility Study Cost-Benefit Analysis (CBA) Methods for Information Gathering Current Business Process/Workflow Functional and Non-Functional Requirements Data Flow Diagram (DFD) (as-is system - Scopes) Requirement Analysis (based on Survey/Interview)	**System Analysis and Design** System Design Physical DFD of the To-Be System (may include partitioning)	
	Database System Boundaries Current Business Rules (Current Business Process/Workflow) Data Requirement Transaction Requirement Planning (Work Breakdown Structure (WBS), Gantt Chart)	**NEW Database** Database Conceptual Design Conceptual Entity-Relationship Model (ERD) Enhanced ERD Data Dictionary	**NEW Database** Database Logical Design Logical ERD Updated Data Dictionary Normalization Updated Data Dictionary Relational Database Schemas (after Normalization) SQL Statements (DDL & DML)
	Appendix Survey/Interview Sample (*if applicable)		
***Note:**	*Discussion for P1 (Proposal) before Submission* *refer to your section lecturer*	*Submission on P2 (draft) Individually* *Iterative process starts from W04* *Might have multiple submissions* *refer to your section lecturer* *refer to your section lecturer*	*Submission on P3 (draft) Individually* *Iterative process starts from W10* *Might have multiple submissions* *refer to your section lecturer* *refer to your section lecturer*